WEEK 4 · RESP-100 BASIC EKG INTERPRETATION

Analyzing EKG Rhythm Strips

Walraven, *Basic Arrhythmias* 8th ed. · Chapter 3 + Appendix C (12-lead review) · Instructor: Michael V Hyland

In one line: every strip gets the same 5-step workup (Regularity → Rate → P waves → PRI → QRS), and the test asks you to fill in exactly those five answers.

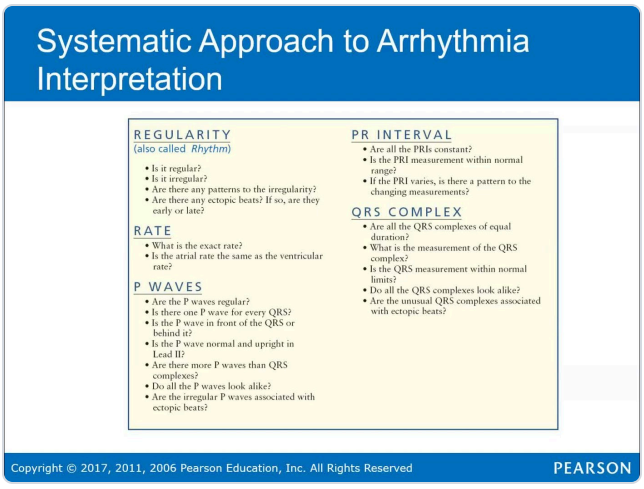
1 · The 5-step method | 2 · Regularity | 3 · Heart rate | 4 · P waves | 5 · PR interval | 6 · QRS width | 7 · Narrow vs wide | 8 · 12-lead review | Worked example | Cheat sheet | Quiz

1 The 5-Step Systematic Approach

Big idea: Don't guess a rhythm by "look". Collect the same five clues, in the same order, every time, then compare them to the rules for each arrhythmia.

1	2	3	4	5
REGULARITY	**RATE**	**P WAVES**	**PRI**	**QRS**
R-R spacing	1500 ÷ boxes or R × 10	upright? uniform?	0.12–0.20 s	≤ 0.12 s = narrow

The five pieces of information you gather from every 6-second strip.



Textbook checklist (slide 3). Each column = questions for one step.

WHAT THE INSTRUCTOR STRESSED · 1:30:03

Test = 40 multiple-choice + **5 six-second strips**. You will **not** have to name the rhythm yet. Under each strip: **Regularity, Rate, P waves, PRI, QRS** = 1 point each (5 points per strip).

GRACE PERIOD · 1:42:00

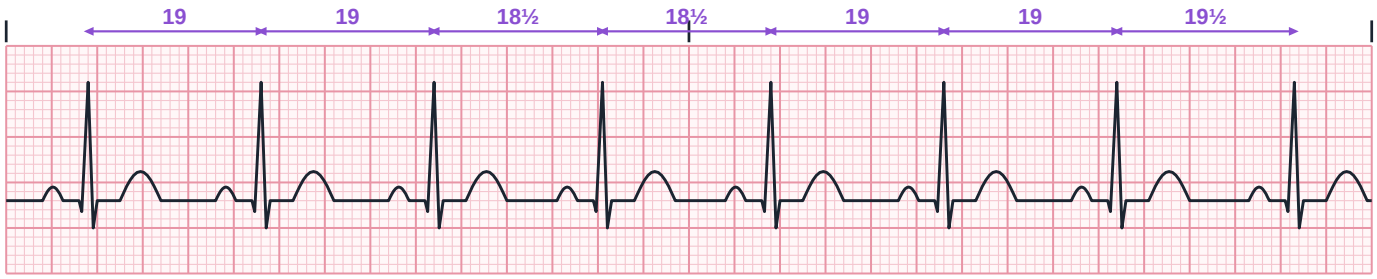
On PRI and QRS he allows **one small box** of difference. If he gets 0.06 and you write 0.08, it's still correct.

MEMORY TRICK

"Really Rapid Pulses Plus QRS"
Regularity · Rate · P waves · PRI · QRS

2 Step 1 – Regularity (R-to-R spacing)

Big idea: Count the **small boxes between each pair of R waves** across the whole strip. If every spacing is within **1½ boxes** of the others, the rhythm is **regular**.

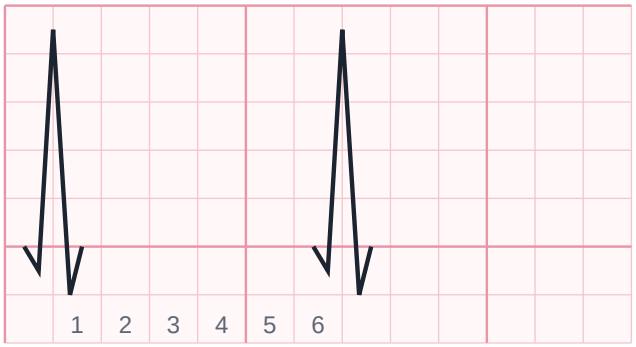


Practice Strip 3.1 (redrawn). The instructor's counts between R waves: 19, 19, 18½, 18½, 19, 19, 19½. Biggest spread = 1 box → **REGULAR**. The tick marks on top are 3 seconds apart.

How he counts (0:37:21)

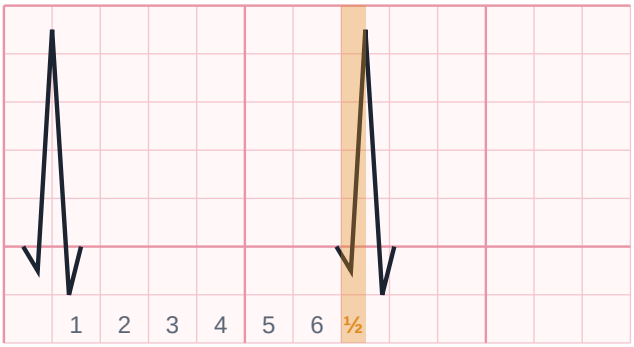
- Count **in fives** using the large boxes: "5, 10, 15…" then add the leftover small boxes.
- **Write the number above the strip** between the two R waves so you never count it twice.
- Do it for **every** R-R pair, not just one.

R wave lands ON a line



6 boxes → count it as a WHOLE box

R wave lands IN the middle



6½ boxes → count it as a HALF box

"Halfsies and wholesies": an R wave that sits on a line counts as a whole box; one that falls between lines counts as a half.

THE 1½-BOX RULE · 0:40:06

All R-R spacings within **1½ small boxes** of each other → **regular**.
Difference **greater than 1½ boxes** → **irregular**.
e.g. 19, 19, 19, 20, 21, 22 → irregular.

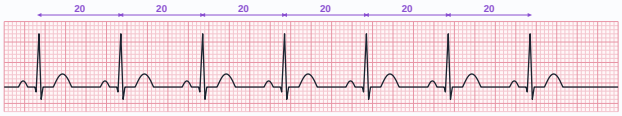
COMMON TRAP

Comparing only the first two beats. A strip can start regular and change halfway, so count **all** the spaces.

2 The 4 Kinds of Regularity

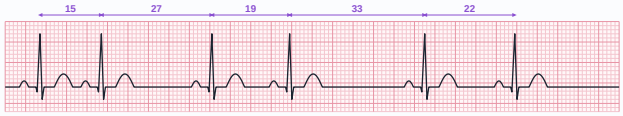
Big idea: The main answers are **regular** or **irregular**. Two more patterns: **regular with ectopic(s)** and **regularly irregular**.

A · Regular



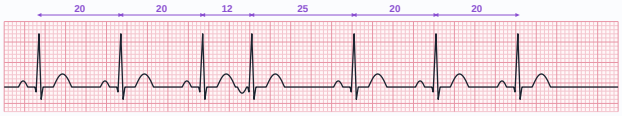
Every spacing the same (±1½ boxes).

B · Irregular



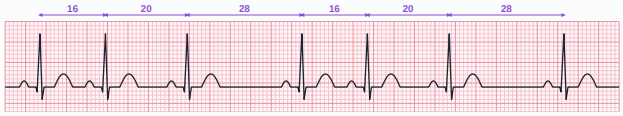
Spacings jump around (15 → 33). No pattern.

C · Regular with 1 ectopic



Regular rhythm interrupted by one **early** beat (different P). Can be 1, 2 or 3 ectopics.

D · Regularly irregular



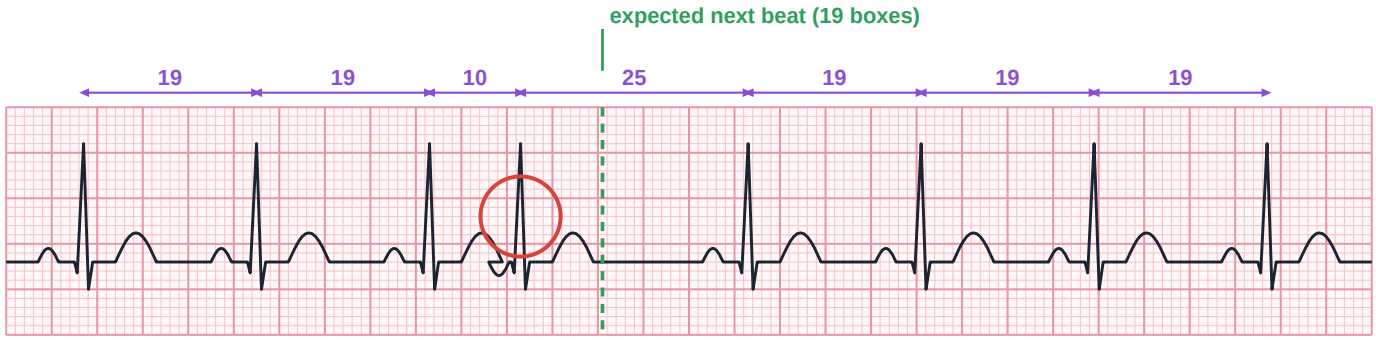
16-20-28 is irregular, but the **same pattern repeats**.

BROKEN RECORD · 0:45:56

Regularly irregular sounds like an oxymoron ("like jumbo shrimp"). Think of a **scratched record** that skips back and replays the same bit: the beats are unevenly spaced, but that uneven group **repeats regularly**. Comes back with heart blocks at the end of the semester.

WHY ECTOPICS HAPPEN · 0:43:39

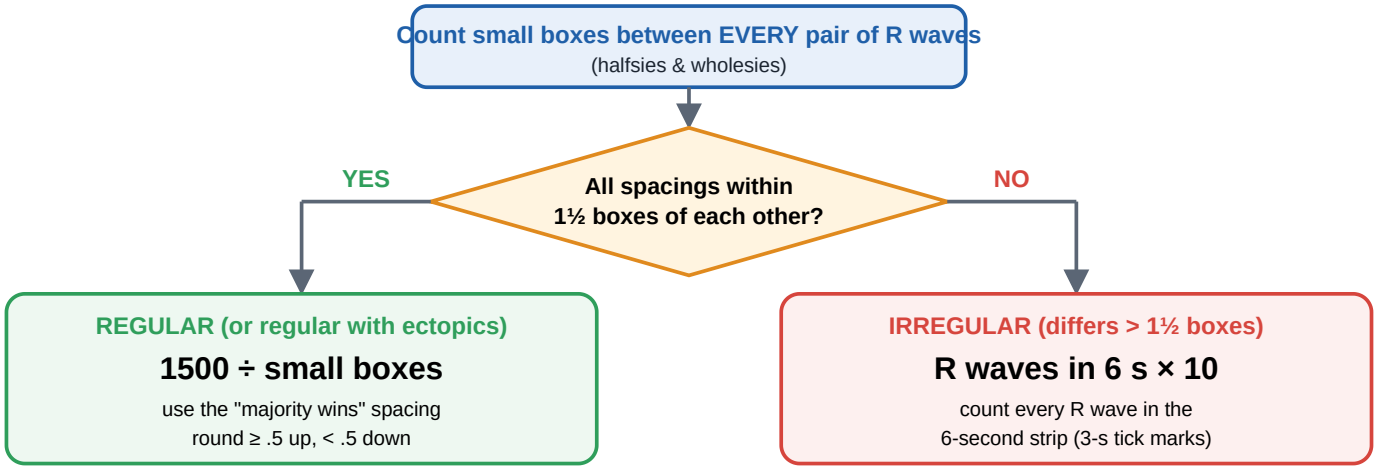
Irritability makes a pacemaker cell fire **early** (premature) before the SA node fires again. Early beats come from the atria (PAC), the junction (PJC) or the ventricles (PVC).



Premature beat: the SA node normally fires every 19 boxes. An irritable spot fires at 10 boxes, which is **earlier** than expected, so it's a premature (ectopic) beat. Its P wave looks different (here inverted).

3 Step 2 – Heart Rate: Only Two Methods

Big idea: Pick the method from the regularity. **Regular → 1500 ÷ small boxes. Irregular → R waves in 6 seconds × 10.**



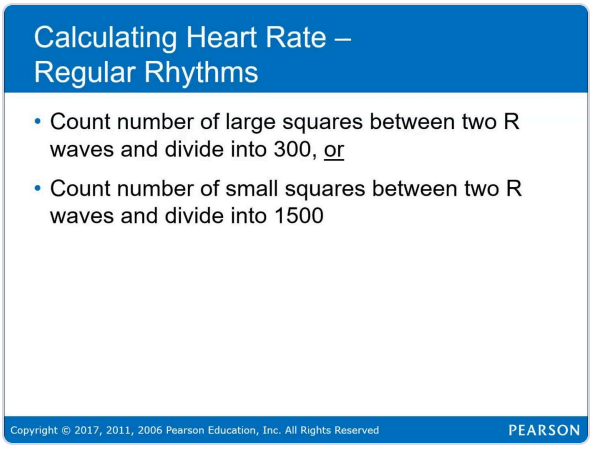
Rate decision tree: regularity decides which formula you are allowed to use.

CROSS IT OUT! · 0:46:44

The slide lists "count **large** squares and divide into 300". The instructor said to **cross that method out**. Counting large boxes isn't accurate enough. Use **small boxes into 1500**.

WHY 1500?

1 small box = 0.04 s, and 60 s ÷ 0.04 s = **1500 small boxes per minute**. Dividing by the boxes per beat gives beats per minute.

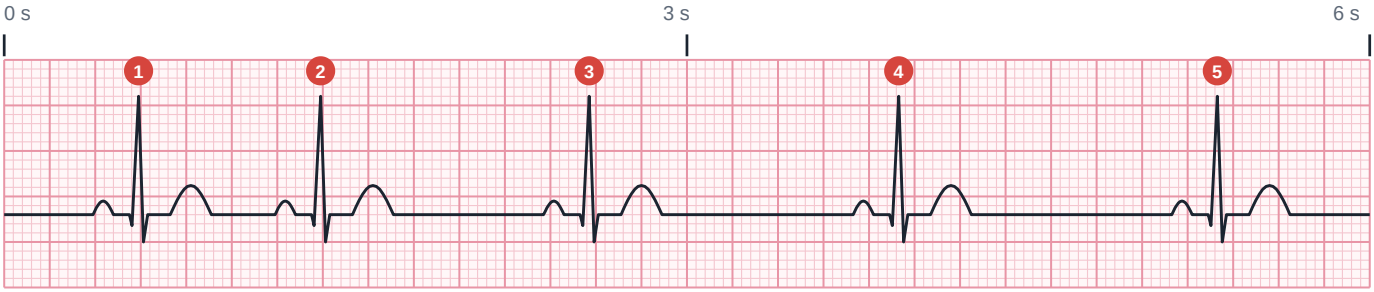


Slide 7: strike out the first bullet (÷ 300).

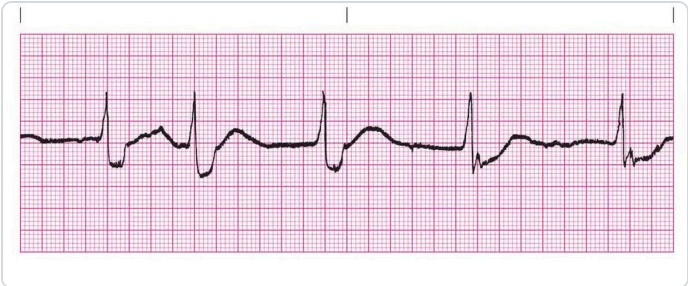
Worked example: Method 1 (regular) on Strip 3.1

Step	What you do	Strip 3.1
1 · Count	Small boxes between each R-R pair	19, 19, 18½, 18½, 19, 19, 19½
2 · Choose	**"Majority wins"**: use the most common spacing	19 (appears 4 times)
3 · Divide	1500 ÷ that number	1500 ÷ 19 = 78.9
4 · Round	≥ 0.5 round up; < 0.5 round down	**79 bpm**

Worked example: Method 2 (irregular): count R waves × 10



Redrawn irregular strip from slide 5 (R-R = 20, 29½, 34, 35 boxes, so it's irregular). Five R waves in 6 s → **5 × 10 = 50 bpm**.



The original slide strip: count the R waves between the first and last 3-second tick (6 s total).

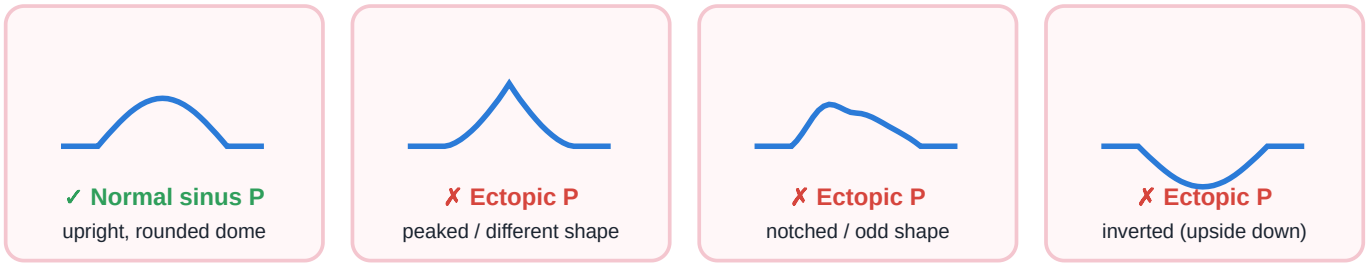
Small boxes R-R	1500 ÷ boxes
10	150 bpm
15	100 bpm
19	79 bpm
20	75 bpm
25	60 bpm
30	50 bpm
37½	40 bpm

WHAT THE INSTRUCTOR STRESSED · 0:48:45

The 1500 method can **only** be used when R-to-R is regular (including regular with 1–3 ectopics). If it's irregular, the spacing varies too much to give an accurate rate, so count R waves × 10. "Those are the only two methods we're going to use in this class."

4 Step 3 – P Waves

Big idea: A normal P wave is **upright** (positive) and **dome-shaped**, and **every P looks the same**. On the test write **"U/U" = Upright / Uniform**.



Morphology = shape. Same shape every time means the same pacemaker every time (the SA node). A different-looking P means a different (ectopic) pacemaker.

Questions to ask (slide 14)

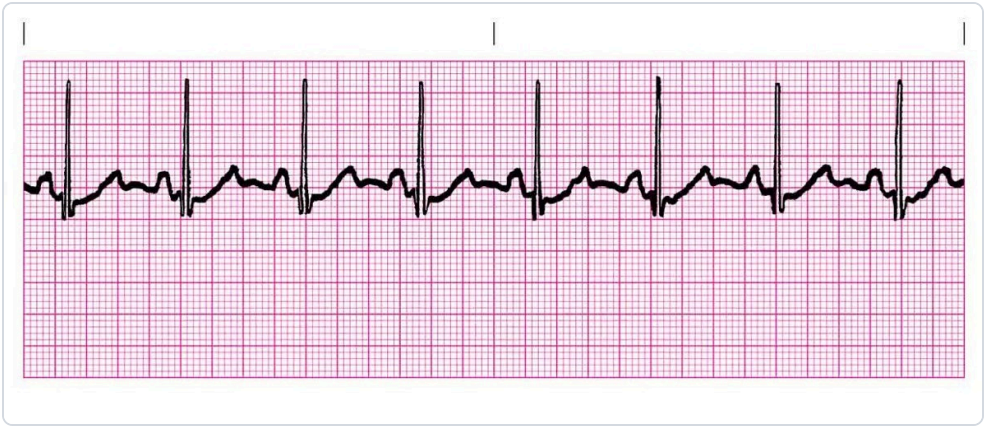
- Are the P waves upright in Lead II?
- Do they all look alike?
- Is there **one P before every QRS**?
- P in front of the QRS, or behind it? (behind = junctional, later)
- More P waves than QRS complexes?
- Any odd P waves that go with early (ectopic) beats?

EXAM ANSWER FORMAT · 1:38:06

"Positively deflected = **upright**. Do they all look the same? Don't split hairs: **uniform**." Write **U/U** or "U and U".

WHY P COMES FIRST

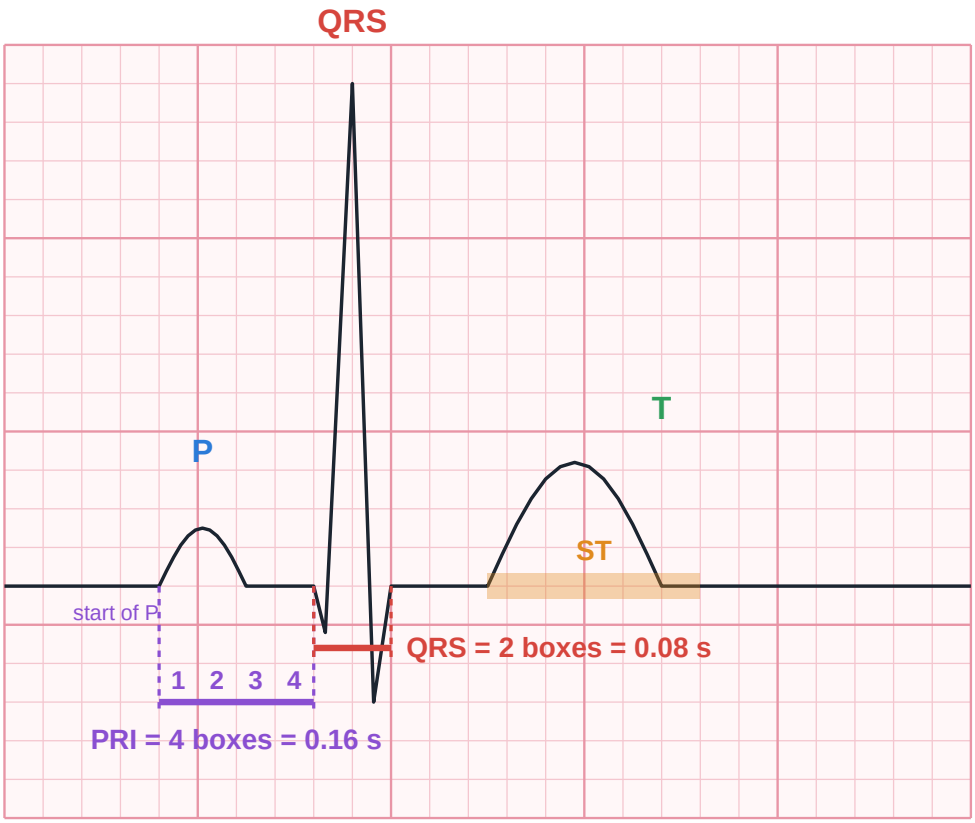
We want the heart to beat **from the top down**: atria first (P), then ventricles (QRS).



Strip 3.1 (slide): find the small rounded bump before each tall spike. All are upright and all look the same → **U/U**.

5 Step 4 – PR Interval (PRI)

Big idea: Measure from the **start of the P wave** to the **start of the QRS**. Normal = **0.12–0.20 s (3–5 small boxes)**, and it should stay the **same across the strip**.



Magnified complex from Strip 3.1: P starts on a line, QRS starts 4 small boxes later → PRI = 4 × 0.04 = **0.16 s** (normal). QRS = 2 boxes = **0.08 s** (narrow).

Small boxes	Seconds
½	0.02
1	0.04
1½	0.06
2	0.08
3	0.12
4	0.16
4½	0.18
5 (1 large)	0.20

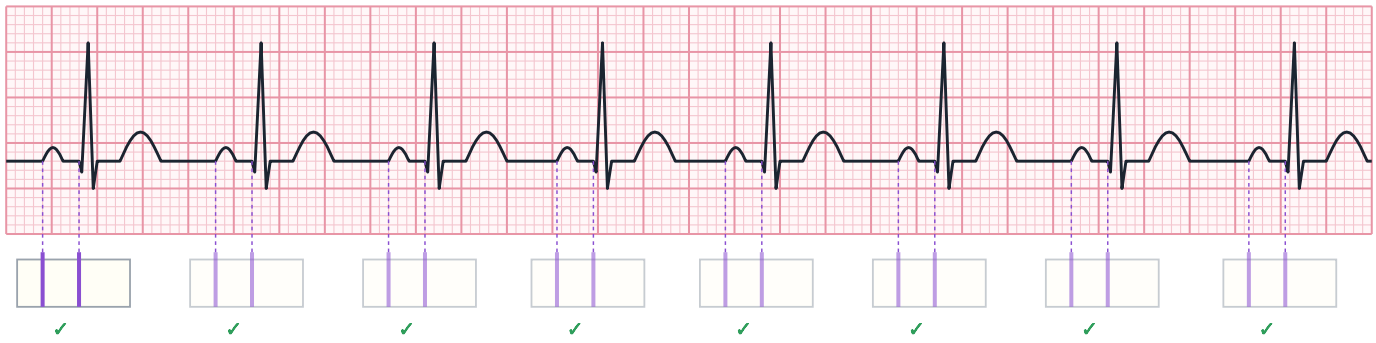
IF THE PRI CHANGES · 0:56:38

A PRI that varies means something is changing: it may be a **different (ectopic) beat** or a **heart block**. For most rhythms the PRI shouldn't change much.

HALF BOXES COUNT

4½ boxes = 0.16 + 0.02 = **0.18 s** (homework strip, 1:40:09).

The scrap-paper "caliper" trick (0:57:28)

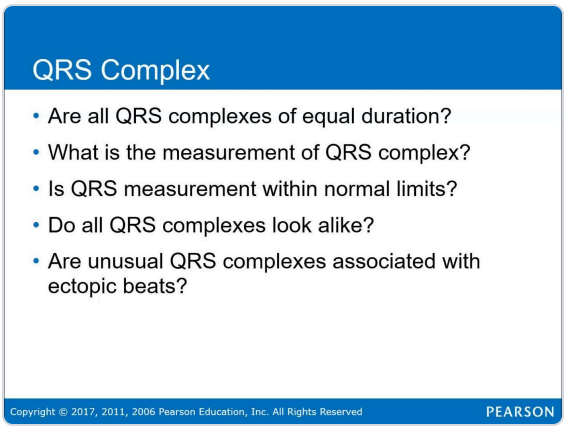


Mark the PRI once on scrap paper, then march the paper across: every PRI fits the same marks → PRI is constant.

The grid can "cloud your vision". Mark the start of P and start of QRS on a blank piece of paper, count the boxes once, then slide it along to check every beat. Do the same for QRS width.

6 Step 5 – QRS Width

Big idea: Measure from where the QRS **leaves** the baseline to where it **returns**. Are they all the same width? Do they all look alike? Is it within normal limits?



Slide 19: the QRS questions.

- Strip 3.1: starts on a line, ends on a line → **2 small boxes = 0.08 s**.
- Homework strip #11: 1½ boxes → **0.06 s** (0.08 also accepted: one-box grace).
- Ectopic beats may have a QRS that looks a bit different.
- March your paper marks across to check that every QRS is the same width.

ROLE OF THE AV NODE · 0:59:12

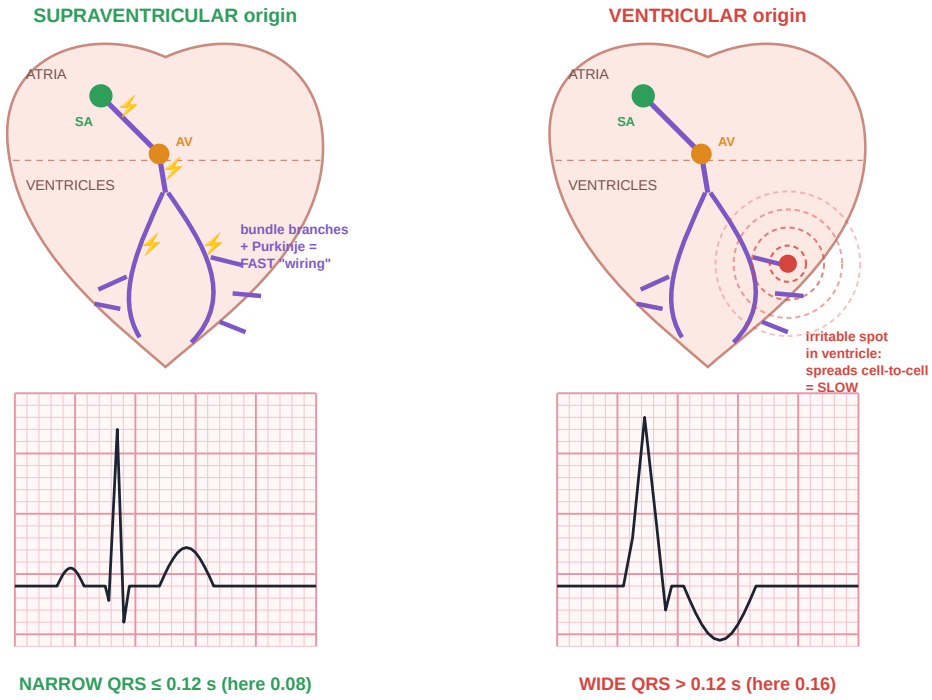
The AV node **"holds"** the impulse so the atria can finish contracting before the ventricles, which **maximizes stroke volume**. That hold is what makes the PRI.

7 Narrow vs Wide QRS: Where Did the Beat Come From?

Big idea: **Narrow QRS (≤ 0.12 s) = supraventricular** (from above the ventricles). **Wide QRS (> 0.12 s) = usually ventricular.** "Supra" = above.

NUMBERS TO KNOW

The instructor's cut-off: **0.12 s or less = narrow/normal**, **greater than 0.12 s = wide**. (Many textbooks write normal as < 0.12 s; follow the lecture's wording on his tests.)



Left: an impulse from the SA node, atria or AV node travels down the fast bundle branches and Purkinje fibres, so the ventricles depolarize quickly and the QRS is narrow. Right: an irritable spot in the ventricle has no wiring to use, so the impulse spreads slowly cell to cell and the QRS is wide.

Supraventricular = SA node, atria, or AV node (incl. bundle of His)

Fast conduction → **narrow** QRS. "Narrow-complex tachycardia" = "supraventricular tachycardia" (same thing).

Wide QRS can mean…

(1) a **ventricular pacemaker** (most of the time), or (2) a supraventricular beat **delayed** in a bundle branch. He said he **won't give you** type (2) this semester.

NOT THE SAME THING: "TOMBSTONES" · 1:08:24

Tombstones are **ST elevation** (S wave runs straight into the T wave) in someone having a heart attack. They are not a wide QRS.

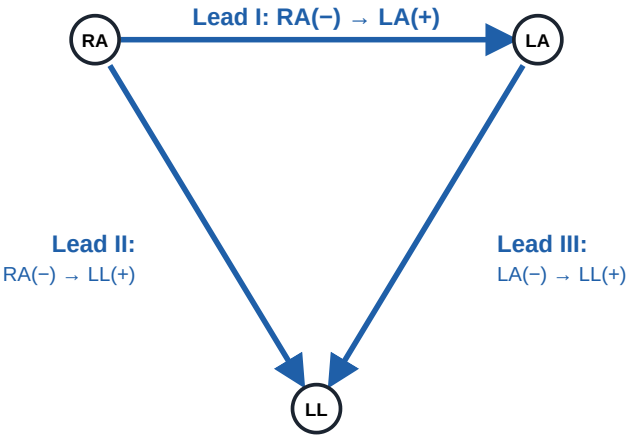
WHAT SLOWS CONDUCTION? · 1:09:09

Coronary artery disease, a past MI, viral infection, and electrolyte imbalance (e.g. high blood potassium stops K^+ leaking out of the cell, so the cell can't recover normally).

8 12-Lead EKG Review (Appendix C)

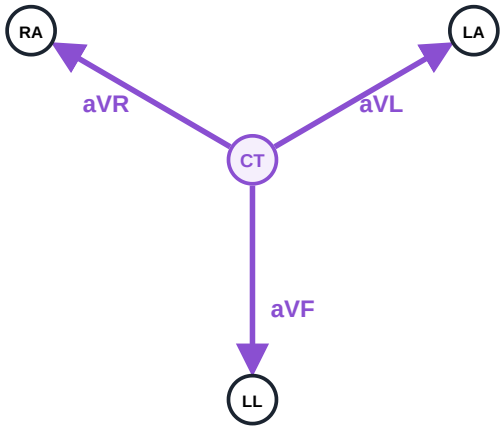
Big idea: 10 electrodes give **12 views** of the heart. Each lead "looks at" one surface, so ST/Q-wave changes in a lead point to the surface (and coronary artery) that's in trouble. **Only 1–2 test questions** (1:32:24).

Standard limb leads (bipolar: − and +)



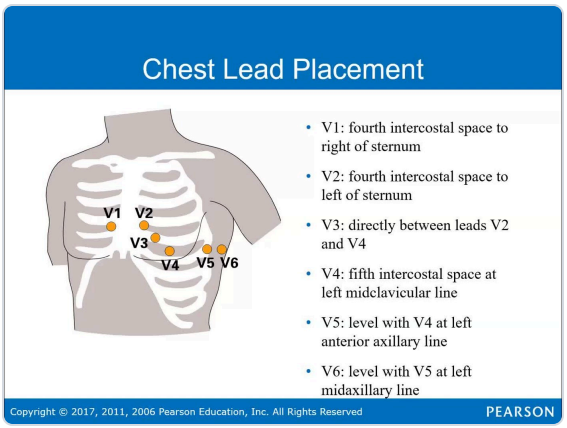
Lead II = most common monitoring lead

Augmented leads (unipolar: + only)



CT = central terminal (neutral centre point). aVR waves are normally inverted.

Frontal-plane leads. A lead = a pair of electrodes on opposite sides of the heart. Current toward + gives an upright deflection; toward − gives a downward one; at right angles gives a flat (isoelectric) line.



Chest (precordial) leads V1–V6, the horizontal plane.

Indications for a 12-lead (slide 30)

- Chest pain or **atypical** chest pain
- Epigastric pain ("heartburn")
- Back, neck, jaw or arm pain without chest pain
- Palpitations · syncope / near-syncope
- Pulmonary edema · exertional dyspnea

Main purpose: diagnose a **myocardial infarction (MI)**. No pain at all = "silent MI".

I Lateral	**aVR** —	**V1** Septal	**V4** Anterior
II Inferior	**aVL** Lateral	**V2** Septal	**V5** Lateral
III Inferior	**aVF** Inferior	**V3** Anterior	**V6** Lateral

Heart surface

- Inferior
- Lateral
- Septal
- Anterior

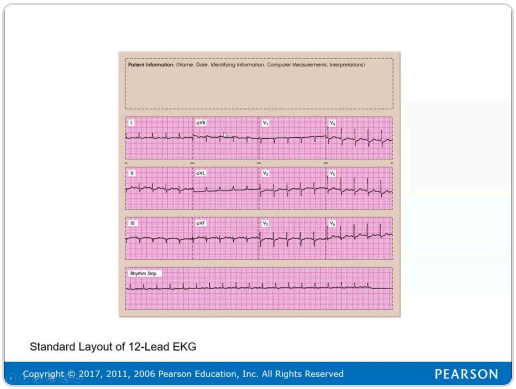
V3–V4 ST elevation = anterior wall MI (left ventricle, LAD)

Rhythm strip (usually Lead II) — full 6 seconds

The top 12 boxes are short "snippets" of each lead.

Colour groups follow the standard textbook lead-to-surface grouping.

Standard 12-lead printout: 3 rows × 4 columns of short snippets, plus a 6-second rhythm strip (usually Lead II) along the bottom.



Slide 39: the real layout. Patient info prints at the top (scan the wristband in clinical).

MI STORY IN BRIEF · 1:13:34

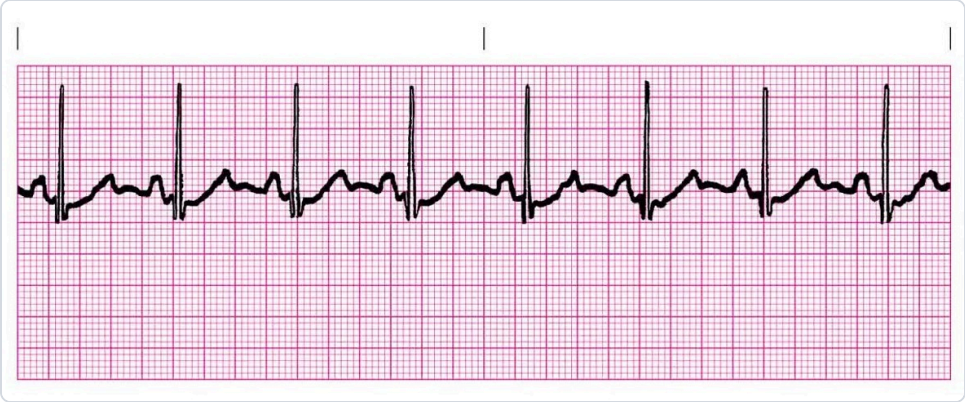
Plaque → cracks → platelets clump → artery blocked → muscle goes **anaerobic** → **angina**. ED care: O_2, aspirin/blood thinner, nitroglycerin, morphine → cath lab (stent) or bypass (CABG); tPA if no cath lab.

CHECK WITH YOUR TEXTBOOK

In lecture the lead-to-wall mapping was described quickly (e.g. lead III called "lateral"). The colours above use the **standard** grouping (II, III, aVF = inferior). Check Appendix C if a test question asks.

★ Putting It Together: The Exam Answer Card

Big idea: On the test, each strip has five blanks underneath. Here is a full answer for Practice Strip 3.1, using the instructor's own numbers.



Practice Strip 3.1, reviewed again at the end of class (1:33:00).

Blank	Answer	How you got it
Regularity	**Regular**	R-R 18½–19½ boxes; spread of 1 box is within 1½
Rate	**79 bpm**	Majority spacing 19 → 1500 ÷ 19 = 78.9 → round up
P waves	**U/U**	Upright (positive) and uniform (all the same shape)
PRI	**0.16 s**	4 small boxes × 0.04, the same on every beat
QRS	**0.08 s**	2 small boxes; narrow → supraventricular

HOMEWORK STRIP #11 (1:35:22)

Counts 19, 19, 19½, 20, 19½, 19, 18½ → spread 18½–20 = 1½ boxes → still **regular**. Rate ≈ **79** (he worked 1500 ÷ 19). P = U/U. PRI = 4½ boxes = **0.18 s**. QRS = 1½ boxes = **0.06 s**.

✓ One-Page Cheat Sheet

Step	How to measure	Normal / rule	Write on test
1 · Regularity	Small boxes between every R-R pair; halfsies & wholesies; write counts on the strip	All within **1½ boxes** = regular; > 1½ = irregular. Also: regular with 1–3 ectopics; regularly irregular (broken record)	Regular / Irregular
2 · Rate	Regular: **1500 ÷ small boxes** (majority wins). Irregular: **R waves in 6 s × 10**	Round ≥ .5 up. **Don't** use 300 ÷ large boxes	___ bpm
3 · P waves	Look before each QRS: shape, direction, one per QRS?	Upright, rounded dome, all alike, one before each QRS	U/U
4 · PRI	Start of P → start of QRS; march paper across	**0.12–0.20 s** (3–5 boxes), constant	0.__ s
5 · QRS	Leaves baseline → returns to baseline	**≤ 0.12 s** narrow = supraventricular; > 0.12 wide = usually ventricular	0.__ s

Grid math

1 small box = **0.04 s**
½ small box = 0.02 s
1 large box = 5 small = **0.20 s**
6-s strip = 30 large boxes
Tick marks = every 3 s

Terms

Ectopic: beat from somewhere other than the SA node
Irritability: cell fires early
Premature: early
Morphology: shape
Supra: above

Grading

5 strips × 5 points
1 point per blank
±1 small box grace on PRI & QRS
No rhythm names needed yet
1–2 Qs on 12-leads

Limb leads

I: RA → LA · II: RA → LL · III: LA → LL
aVR, aVL, aVF: unipolar from the central terminal
aVR normally upside down
Lead II = usual monitoring lead

12-lead essentials

10 electrodes → 12 leads
Purpose: find an MI (ST/Q changes)
V3–V4 ST elevation = anterior wall (LV, LAD)
Rhythm strip at bottom = Lead II

? Self-Quiz

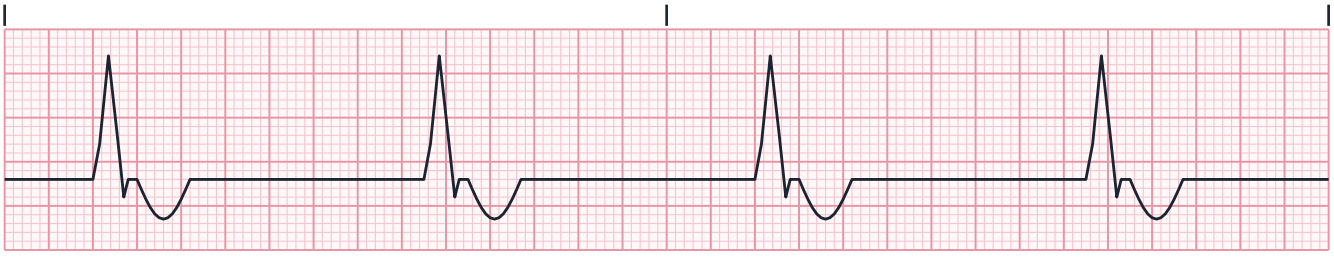
1. List the five pieces of information you gather from every rhythm strip, in order.
2. R-R counts across a strip are 22, 22, 23, 21½, 22, 22½. Regular or irregular? Why?
3. Why did the instructor tell you to cross out the "300 ÷ large boxes" method?
4. When are you **not** allowed to use the 1500 method? What do you use instead?
5. An R wave lands halfway between two small-box lines. How do you count that box?
6. Gather all five pieces of information for Strip A.



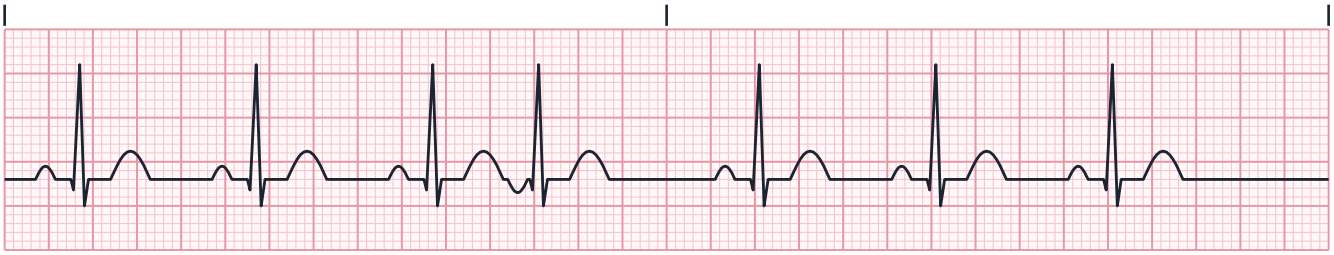
7. Strip B: Is it regular? What is the heart rate, and which method did you use?



8. Strip C: Measure the QRS. Narrow or wide? Where does this rhythm most likely come from, and why? What is the rate?



9. Strip D: Describe the regularity in the instructor's words. What is unusual about the 4th beat?



10. What does "regularly irregular" mean? Use the instructor's analogy.

11. A PRI measures 4½ small boxes. What is it in seconds, and is it normal?
12. Why does an impulse from the SA node produce a narrow QRS, but one from an irritable ventricular cell produce a wide QRS?
13. What does the AV node "hold", and why is that useful?
14. Name the electrodes for leads I, II and III. Which lead is normally all upside down?

Answers

1. Regularity, Rate, P waves, PRI, QRS.
2. **Regular**: range 21½–23 = 1½ boxes, which is not more than 1½.
3. Counting large boxes isn't accurate. Small boxes into 1500 is the accurate method.
4. When R-R is **irregular**. Count R waves in the 6-second strip × 10.
5. As a **half** box (halfsies); on a line = whole (wholesies).
6. Regular (25 boxes each); 1500 ÷ 25 = **60 bpm**; P = U/U; PRI = 5 boxes = **0.20 s** (upper limit of normal); QRS = 2 boxes = **0.08 s**.
7. **Irregular** (spacings 14–26). Count R waves: 7 × 10 = **70 bpm** (6-second method).
8. QRS = 4 boxes = **0.16 s → wide**. No P waves; wide = most likely **ventricular**: the impulse spreads cell to cell (slow) instead of down the fast bundle branches. Regular at 37½ boxes → 1500 ÷ 37.5 = **40 bpm**.
9. **Regular with one ectopic**. Beat 4 comes early (12 boxes instead of 20 = premature) and has a different (inverted) P wave, so it comes from an irritable ectopic pacemaker. Rate = 1500 ÷ 20 = 75 bpm.
10. The spacing is irregular, but the irregular group of beats repeats over and over, like a **scratched/broken record** replaying the same part of a song.
11. 4 × 0.04 + 0.02 = **0.18 s**; normal (0.12–0.20).
12. SA impulses travel down the fast, nerve-like bundle branches and Purkinje fibres, so the ventricles contract quickly (narrow). A ventricular focus has no fast pathway to use, so it spreads slowly cell to cell (wide).
13. It holds the impulse until the ventricles can receive it, letting the atria finish contracting first, which **maximizes stroke volume**.
14. I: RA(−) → LA(+); II: RA(−) → LL(+); III: LA(−) → LL(+). **aVR** is normally inverted.

Based on the Week 4 lecture (Chapter 3 + Appendix C) of RESP-100. Strips labelled "redrawn" or "Strip A–D" are illustrations drawn to the standard 25 mm/s grid.